Taxing the Rich II: Entrepreneurs and Ideas

Lutz Hendricks

2026-04-21

UNC Chapel Hill

Entrepreneurs

Basic Idea

Entrepreneurs invest in building up their businesses.

- If others earn rents from high productivity (workers?), don't tax them.

Model generation 1:

- Just stick progressive taxes into a Cagetti & De Nardi (2006) model of wealth distribution.
- E.g., İmrohoroğlu, et al. (2023)

Model generation 2:

- Entrepreneurs create ideas that benefit the whole economy.
- Jones (2022)

İmrohoroğlu, et al. (2023): Model Ingredients

Essentially a Cagetti & De Nardi (2006) model with progressive taxes:

- Stochastic aging (two "ages": young and old, as in Castañeda et al.)
- Two sectors: corporate (Cobb-Douglas) and entrepreneurial
- Entrepreneurs run a DRS technology: $f(k,n) = \theta\left(k^{\gamma}(l+n)^{1-\gamma}\right)^{\nu}, \nu < 1$
  - θ: entrepreneurial ability (stochastic)
  - Choose own labor l, hire workers n, invest capital k
- Occupational choice: each period, young agents choose worker vs. entrepreneur
- Borrowing constraint: $k \leq (1+d)a$ (collateral-based)

Key difference from KK: taxes fall on **total income** (labor + capital), not just labor earnings.

Results

Revenue- and welfare-maximizing top marginal rate on the top 1%: **55%**

But: when only **labor income** is taxed (as in KK), the optimal rate rises to **80%**.

Intuition:

- Entrepreneurs earn mostly capital income
- Taxing capital income distorts business investment and capital allocation
- This is the main margin that lowers the optimal rate
- Without it, the result looks like Kindermann-Krueger

Discussion

Not much has changed relative to the superstar model.

But: **entrepreneurs don't do anything useful**.
Their businesses benefit only themselves.
So why not tax them?

Jones (2022): Taxing Innovators

Motivation

Question

What if top earners create ideas that benefit the whole economy?

The obvious answer:

- subsidize R&D
- tax high incomes

This is not implementable.

Much innovation happens outside formal R&D:

- Walmart, Uber, Goldman Sachs report zero R&D
- Cannot be targeted by R&D tax credits
- The income tax is the only instrument available

Motivation

Key insight:
Entrepreneurs generate knowledge spillovers.

Tax the rich

$\rightarrow$ reduce R&D effort
$\rightarrow$ reduce other firms' productivities
$\rightarrow$ large welfare losses

Jones: Model Structure

Reference: Jones (2022)

An R&D growth model of increasing varieties - with a twist:

- **R&D uses effort**

That is the decision that taxes distort.

With free entry as the distorted margin:

- Lower profits after innovation.
- The interest rate must fall to restore zero profits.
- The effect of taxes becomes very indirect.

Jones: Demographics

Population grows at rate n.

Occupational choice is fixed:

- $N_t = \overline{N} e^{nt}$ workers
- $M_t = \overline{M} e^{nt}$ managers
- $S_{at} = \overline{S}_a e^{nt}$ entrepreneurs

Taxes distort effort, not occupational choice.

All agents live forever

Jones: Preferences

$$\max_{c,e} \int_0^\infty \left[\phi \log c - \frac{\epsilon}{1+\epsilon} e^{(1+\epsilon)/\epsilon}\right] \exp(-\delta t) dt$$

e is effort

- managers and innovators only

Taxes reduce effort.

- That is the only distorted margin in this model.

Jones: Technologies - Final Goods

$$Y_t = \tilde{M}_t^{\psi} \int_0^{A_t} x_{it}^{1-\psi} di$$

- uses managers $\widetilde{M}$ and intermediates
- $\widetilde{M} = \mathbb{E}[\theta e] \cdot M$
- θ = ability
- e = effort (a choice)

All growth comes from increasing varieties A_t.

Note: Ability heterogeneity does not do anything in this model.

Jones: Intermediate goods

$$x_{it} = \ell_{it}$$

One unit of worker labor per unit of output

- Marginal cost = worker wage w_t

Each variety i produced by a monopolist with a patent

- Monopoly markup over marginal cost

Jones: Applied R&D

Aggregate idea production function:

$$\dot{A}_t = \bar{a}\tilde{S}_{at}^{\lambda} A_t^{\varphi_a} B_t^{\beta_a}$$

- $\tilde{S}_{at} = S_{at} \cdot \mathbb{E}[\theta e]$: effective entrepreneurial input (talent $\times$ effort)
- A_t: existing stock of applied ideas
- B_t: stock of basic research

Why two types of R&D?

- Sidestep the objection "just tax income and subsidize R&D"
- Applied R&D is not taxable.

Jones: Basic R&D

$$\dot{B}_t = \bar{b} S_{bt}^{\lambda} B_t^{\varphi_b}$$

with $\varphi_b < 1$

Government chooses number of basic researchers S_{bt}.

They come out of worker pool: $N_t = L_t + S_{bt}$

Growth of B_t is effectively exogenous.

Looking ahead: Balanced growth rate of B_t must satisfy

$$\lambda n + (\varphi_b - 1) g(b) = 0$$

Semi-endogenous growth

Reproducible factors:

$$\dot{A}_t = \bar{a}\tilde{S}_{at}^{\lambda} A_t^{\varphi_a} B_t^{\beta_a}$$

$$\dot{B}_t = \bar{b} S_{bt}^{\lambda} B_t^{\varphi_b}$$

Assume $\varphi_a + \beta_a, \varphi_b < 1$

- diminishing returns to knowledge

$\rightarrow$ semi-endogenous growth

Jones: Resource Constraints

Labor

Workers supply labor to intermediate goods production or become basic researchers.

$$N_t = L_t + S_{bt}$$

- $L_t = \int_0^{A_t} \ell_{it} di$
- Entrepreneurs (S_{at}) and managers (M_t) are separate, exogenous populations.

Goods

$$Y_t = C_t$$

All output is consumed.
Workers make no choices.

Two-bracket tax schedule:

- Income below $\bar{w}$: taxed at base rate τ_0
- Income above $\bar{w}$: taxed at top marginal rate τ

All entrepreneurs and managers earn above $\bar{w}$.
Basic researchers and workers earn below $\bar{w}$.

Government budget constraint:

$$T = \tau_0 Y + (\tau - \tau_0)\underbrace{[\rho Y - \bar{w}(S_a + M)]}_{\text{rich, above } \bar{w}} = \Omega Y$$

- ρ: income share of entrepreneurs and managers
- Ω: required government spending as a share of GDP

Jones: Entrepreneur's Problem

The entrepreneur is an employee of the innovating firm.

- earns wage w_t.

$$\max_{c,e} \int_0^\infty \left[\phi \log c - \frac{\epsilon}{1+\epsilon} e^{(1+\epsilon)/\epsilon}\right] \exp(-\delta t) dt$$

Budget constraint (continuous time):

$$\dot{v} = (r-n)v + \underbrace{\bar{w}(\tau - \tau_0) + w_t \theta e(1-\tau)}_{\text{net earnings}} + R - c$$

- $w_t \theta e$: earnings from innovation (proportional to talent $\times$ effort)
- R: lump-sum rebate (depends on θ)

Similar for managers (with wage w_m)

Jones: Effort FOC from the Hamiltonian

Current-value Hamiltonian:

$$\mathcal{H} = \phi \log c - \frac{\epsilon}{1+\epsilon} e^{(1+\epsilon)/\epsilon} + \mu[(r-n)v + \bar{w}(\tau - \tau_0) + w_t \theta e(1-\tau) + R - c]$$

First-order conditions:

- $\partial \mathcal{H} / \partial c = 0$: $\quad \mu = \phi / c$
- $\partial \mathcal{H} / \partial e = 0$: $\quad e^{1/\epsilon} = \mu w_t \theta (1-\tau)$
- Costate:

$$\dot{\mu} = (\delta - r + n)\mu$$

Jones: Optimal Effort

Combining the two intratemporal FOCs:

$$e_t = \left(\frac{\phi w_t \theta (1 - \tau)}{c_t} \right)^{\epsilon}$$

Effort still depends on $c_t \rightarrow$ need to specify the rebate $R(\theta)$.

Jones: Why a Rebate is Needed

On the BGP, $g(c(\theta)) = g(w)$.

Budget constraint:
present value of consumption = present value of earnings + present value of rebate

Without a rebate ($R = 0$), consumption is proportional to after-tax earnings:

$$c_t(\theta) \propto w_t \theta e(1-\tau)$$

Substitute into the FOC

$$e^{1/\epsilon} = \frac{\phi w_t \theta (1-\tau)}{c_t}$$

$$\Rightarrow e = \text{constant}$$

Intuition

We wrote down a model with a balanced growth path.

Wages grow over time.

Substitution effect: $c \uparrow$ and $e \uparrow$

- Substitute from leisure to consumption.

Income effect: $c \uparrow$ and $e \downarrow$

Preferences consistent with balanced growth imply that income and substitution effects must cancel.

But then taxes do not affect effort.

Why does Effort Respond to Taxes?

The rebate $R(\theta)$ rises with individual tax payments.

Now we have an additional income effect:

- Higher tax increases the rebate.
- The income effect reduces effort.

The Rebate

Assumption: the rebate function is set such that

$$e^* = [\phi(1-\tau)^\alpha]^{\epsilon/(1+\epsilon)}$$

Effort depends on the keep rate $(1-\tau)$, **not on talent θ**.

α parameterizes how the rebate varies with the tax rate

- $\alpha = 0$: rebate is constant and taxes don't matter
- $\alpha = 1$: full rebate; only substitution effects

Jones works out some of the details…

Jones: Effort and the Elasticity

$$e^* = [\phi(1-\tau)^\alpha]^{\epsilon/(1+\epsilon)}$$

Elasticity of effort wrt the keep rate:

$$\zeta_u \equiv \frac{d \log e^*}{d \log(1-\tau)} = \frac{\alpha\epsilon}{1+\epsilon}$$

- $\alpha = 1$: ζ_u equals the Frisch elasticity $\epsilon/(1+\epsilon)$.
  - Frisch elasticity: response of effort to the keep rate, holding marginal utility of consumption (u_c) constant.
- $\alpha < 1$: income effect lowers $\zeta_u \to$ taxes distort less.

ζ_u is the elasticity calibrated in Tables 2–4.

Jones: R&D Firms and Externalities

The aggregate idea production function is

$$\dot{A}_t = \bar{a}\tilde{S}_{at}^{\lambda} A_t^{\varphi_a} B_t^{\beta_a}$$

But each individual R&D firm takes the productivity of the idea sector as given:

$$\dot{A}_t = \bar{v}_t \tilde{S}_{at}$$

with $\bar{v}_t = \bar{a}\tilde{S}_{at}^{\lambda-1} A_t^{\varphi_a} B_t^{\beta_a}$

Externalities:

- Duplication ($\lambda < 1$): doubling all researchers less than doubles new ideas
- Knowledge spillovers (φ_a, β_a): the existing stocks A_t and B_t raise productivity, but no firm internalizes its contribution to A_t

Jones: R&D Firms and Externalities

The firm's problem:

$$\max_{\tilde{S}_a} P_{At}\bar{v}_t\tilde{S}_{at} - w_{st}\tilde{S}_{at}$$

taking the price of ideas P_{At}, productivity $\bar{v}_t$, and wage w_{st} as given.

Free entry requires $P_{At}\bar{v}_t = w_{st}$.

Jones: Balanced Growth Path

By symmetry ($x_{it} = x_t$ for all i), output simplifies to:

$$Y_t = A_t^{\psi} \tilde{M}_t^{\psi} L_t^{1-\psi}$$

It can be shown that

$$A_t^* = \left(\nu_a \tilde{S}_{at} S_{bt}^{\beta}\right)^{\gamma/\psi}$$

where $\tilde{S}_{at} \equiv \mathbb{E}[\theta e^v] S_{at}$

Jones: Balanced Growth Path

Substituting into the production function:

$$Y_t^* = \left(\nu_a \mathbb{E}[\theta e^v] S_{at} S_{bt}^{\beta}\right)^{\gamma} (\mathbb{E}[\theta e^m] M_t)^{\psi} L_t^{1-\psi}$$

where

- $\gamma \equiv \psi\lambda/(1-\varphi_a)$ and
- $\beta \equiv \beta_a/(1-\varphi_b)$.

Growth rate of output per person:

$$g_y = \gamma(1+\beta)n$$

- semi-endogenous (driven by population growth).

Jones: Two Channels from Taxes to Output

Rewrite steady-state output to highlight the role of effort:

$$Y \propto (e^v)^\gamma (e^m)^\psi$$

Only effort levels e^v and e^m depend on τ.

Elasticity of output with respect to the keep rate:

$$\eta_{Y,1-\tau} \equiv \frac{d \log Y}{d \log(1-\tau)} = \gamma \zeta^v + \psi \zeta^m$$

where ζ^v and ζ^m are the effort elasticities of entrepreneurs and managers.

Two channels from taxes to GDP: $\eta_{Y,1-\tau} = \gamma\zeta^v + \psi\zeta^m$

Manager channel ($\psi\zeta^m$):

- Managers enter **inside** the CRS part of the production function
- ψ competes with other factor shares $\rightarrow$ bounded, small (≈ 0.15)
- This is essentially the standard labor supply channel

Idea channel ($\gamma\zeta^v$):

- Ideas enter **outside** the CRS part (because ideas are nonrival)
- γ is **not part of a sum that equals one** $\rightarrow$ can be large
- This is the new force: taxing entrepreneurs slows innovation, which lowers **everyone's** income

Jones: Revenue-Maximizing Top Rate

Scenario	Revenue max	Welfare max
$\gamma = 0$ (no ideas)	92%	76%
$\gamma = \frac{1}{8}$	91%	57%
$\gamma = \frac{1}{2}$ (baseline)	65%	9%
$\gamma = 1$ (strong ideas)	52%	–43%

$(\zeta_u = 0.2, \Omega = 0.2, \alpha = 1)$

Revenue maximization and welfare maximization diverge sharply.

With $\gamma = \frac{1}{2}$: moving from $\tau = 0.75$ to the welfare optimum raises GDP $\approx$ 15%.

Jones: Discussion

Take-away message:

When top earners create externalities, taxing them is very costly.

Limitation:

- The strength of the externality paramater γ is not known.
- Cannot estimate "the" optimal tax rate.

Jones: Discussion

This is a **proof-of-concept paper**:
Here is a non-crazy model where taxing innovation is a bad idea.

> The point is that the discovery of nonrival ideas by entrepreneurs appears to have a first-order effect on the calculation and should be included in future work.
>
> — Jones (2022) p. 2265

An important open issue:
Can the magnitudes be pinned down better?

Taking Stock

The Big Picture

Paper	Source of top earnings	Key margin	Optimal rate
Kindermann-Krueger	Luck	Labor supply	79%
Guner-Ventura	Persistent ability	Labor supply	37%
Badel-Huggett-Luo	HC investment	Skill investment	49%
İmrohoroğlu et al.	Business income	Capital alloc.	55%
Jones	Innovation	Innovation effort	$\approx$ 22%

Open question: How to model the distribution of earnings?

References

References

- Badel, A., M. Huggett, and W. Luo (2020). "Taxing Top Earners: A Human Capital Perspective." *The Economic Journal*, 130(629), 1200–1225.
- Cagetti, M. and M. De Nardi (2006). "Entrepreneurship, Frictions, and Wealth." *Journal of Political Economy*, 114(5), 835–870.
- Castañeda, A., J. Díaz-Giménez, and J.-V. Ríos-Rull (2003). "Accounting for the U.S. Earnings and Wealth Inequality." *Journal of Political Economy*, 111(4), 818–857.
- Guner, N., M. Lopez-Daneri, and G. Ventura (2016). "Heterogeneity and Government Revenues: Higher Taxes at the Top?" *Journal of Monetary Economics*, 80, 69–85.
- İmrohoroğlu, A., C. Kumru, J. Lian, and A. Nakornthab (2023). "Revisiting Taxes on High Incomes." *Review of Economic Dynamics*, 51, 1159–1184.
- Jones, C. I. (2022). "Taxing Top Incomes in a World of Ideas." *Journal of Political Economy*, 130(9), 2227–2274.
- Kindermann, F. and D. Krueger (2022). "High Marginal Tax Rates on the Top 1%? Lessons from a Life-Cycle Model with Idiosyncratic Income Risk." *American Economic Journal: Macroeconomics*, 14(2), 319–366.